PH 101
Tutorial 3
Date: 22/08/2023

Q.1-Q.5 are to be discussed in the Tutorial class.

1. Which of the following forces are conservative? If conservative, find the potential energy $V(\vec{r})$. $(a, b, c$ are constants)

   (a) $F_x = ayz + bx + c, F_y = axz + bz, F_z = axy + by.$

   (b) $F_x = -ze^{-x}, F_y = \ln z, F_z = e^{-x} + \frac{y}{z}$

   (c) $\vec{F} = \frac{a\hat{r}}{r}$

2. A particle of mass m moving in one dimension has potential energy $V(x) = V_0[2\left(\frac{x}{a}\right)^2 - \left(\frac{x}{a}\right)^4]$ where V_0 and a are positive constants.

   (a) Find the force $F(x)$ which acts on the particle.
   (b) Sketch $V(x)$. Find the positons of stable and unstable equilibrium.
   (c) What is the angular frequency ω of oscillations about the point of stable equilibrium?
   (d) What is the minimum speed of the particle must have at the origin to escape to infinity?

3. A particle of mass m moves in one dimension along the positive x axis. It is acted on by a constant force directed toward the origin with magnitude B, and an inverse-square law repulsive force with magnitude A/x^2.
   (a) Find the potential energy function $V(x)$.
   (b) Sketch the energy diagram for the system when the maximum kinetic energy is $T_0 = \frac{1}{2}mv_0^2$
   (c) Find the equilibrium position, x_0.

4. Consider the function $F = \left(\frac{dy}{dx}\right)^2$, where $y(x) = x$. Add to $y(x)$ the function $\eta(x) = \sin x$, **find** $S(\alpha)$ between the limits of $x = 0$ and $x = 2\pi$. Show that the stationary value of $S(\alpha)$ occurs for $\alpha = 0$.

5. Consider a ray of light traveling in a vacuum from point P_1 to P_2 by way of the point Q on a plane mirror, as in Figure 1. Show that Fermat's principle

implies that, on the actual path followed, Q lies in the same vertical plane as P_1 and P_2 and obeys the law of reflection, that $\theta_1 = \theta_2$.

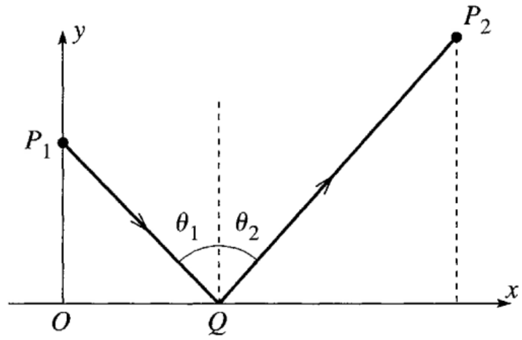

Fig. 1

Practice Problems

6. A particle P of mass *m* moves on the axis Oz under the gravitational attraction of a uniform circular disk of mass *M* and radius *a*. It can be shown that the force field F(z) acting on P is given by

$$F = -\frac{2mMG}{a^2}\left[1 - \frac{z}{(a^2+z^2)^{1/2}}\right] \qquad (z > 0)$$

Find the corresponding potential energy V(z) for z > 0. Initially P is released from rest at the point z= $4a/3$. Find the speed of P when it hits the disk

7. A ray of light travels from point P_1 in a medium of refractive index n_1 to P_2 in a medium of index n_2, by way of the point Q on the plane interface between the two media, as in Figure 2. Show that Fermat's principle implies that, on the actual path followed, Q lies in the same vertical plane as P_1 and P_2 and obeys Snell's law, that $n_1 sin\theta_1 = n_2 sin\theta_2$

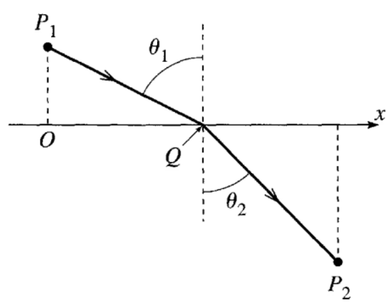

Fig. 2